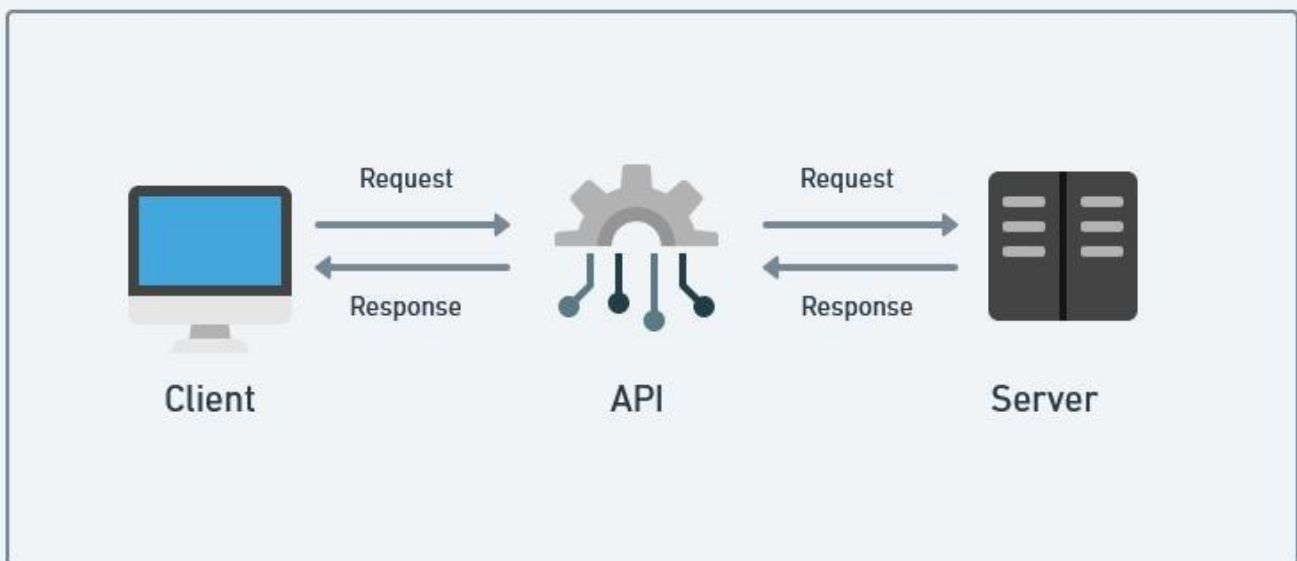
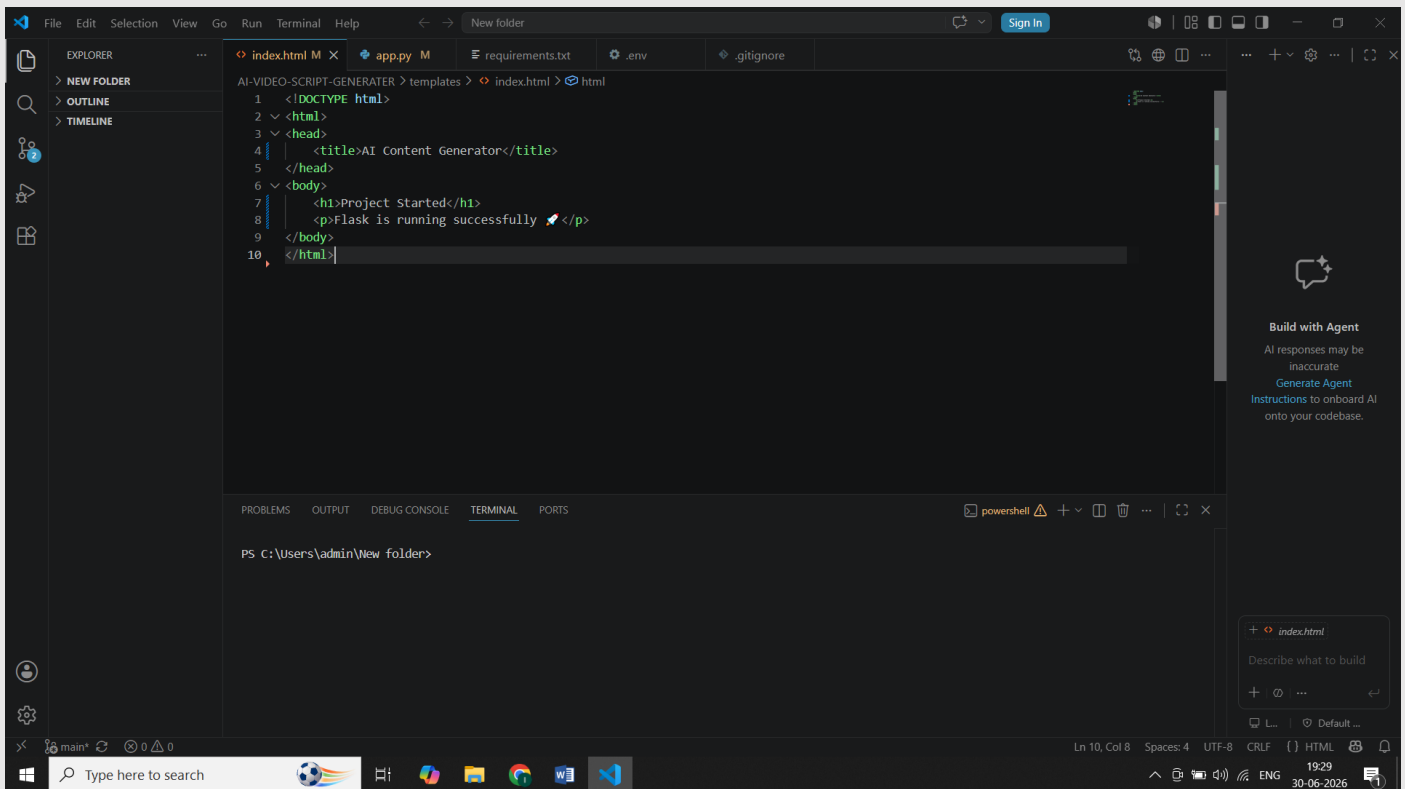
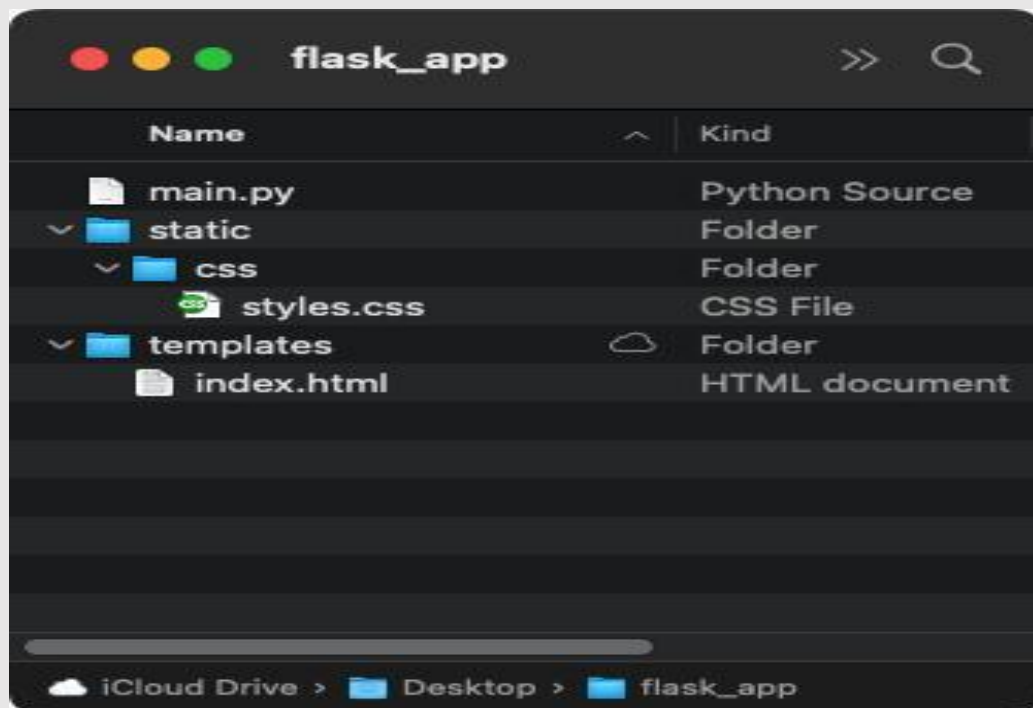


TECHNOLOGY STACK

DATE	30 JUNE 2026
TEAM ID	KANAK DHINDHWAL , ISHA SAINI , BALJEET KAUR
PROJECT NAME	AI POWERED VIDEO SCRIPT GENERATOR
MAXIMUM MARKS	

VS CODE :- VS Code is a code editor where you write and manage your project code.





To build this project in a simple, efficient, and scalable way, we are using a lightweight full-stack approach where Python handles the backend logic and AI integration, while basic web technologies are used for the user interface

1. Frontend (User Interface)

The frontend is designed to be simple and easy to use so that any user can generate scripts without technical knowledge.

- **HTML** – To create the structure of the web pages (input form, buttons, output section)
- **CSS** – To design and style the interface for a clean and user-friendly look
- **JavaScript (optional/basic use)** – For minor interactions like button clicks or dynamic updates

🔗 Purpose: Provides a simple form where users enter topic, audience, tone, and platform.

2. Backend (Core Logic)

The backend handles user input, processes it, and communicates with the AI model.

- **Python** – Main programming language for backend development
- **Flask Framework** – Lightweight web framework used to connect frontend with backend logic
- **Request Handling (Flask routes)** – To receive form data and send responses back to the UI

🔗 Purpose: Acts as the brain of the application.

3. AI Integration Layer

This is the most important part of the project where the actual script generation happens.

- **Groq API / GPT API (LLM Models like LLaMA / Mixtral / GPT)**
Used to generate high-quality video scripts based on user input.
- **Prompt Engineering Layer**
 - Combines user inputs (topic, tone, platform, audience)
 - Converts them into structured AI prompts
 - Ensures output is formatted as a proper video script

☞ Purpose: Converts simple user inputs into professional-level scripts using AI.

4. Development Environment

- **VS Code** – Used for writing and managing code
 - **Python 3.x** – Runtime environment
 - **Pip** – For installing required libraries like Flask and API clients
 - **Virtual Environment (venv)** – To manage project dependencies cleanly
-

5. API & Communication

- **REST API calls** – Used to communicate with Groq/GPT servers
 - **JSON format** – Used for sending input data and receiving AI-generated script
-

6. Optional Enhancements (Future Scope)

These are not required for the current version but can improve the project later:

- **Database (SQLite/MySQL)** – To store generated scripts
- **Authentication system** – For user login and history
- **Cloud deployment (Render / Vercel / PythonAnywhere)**
- **Export options (PDF / Word download of scripts)**